ELK 336E
HOMEWORK #2

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1)

$|0\rangle$ — $\boxed{H}$ — $\boxed{U_\varphi}$ — $\boxed{H}$ — $|\chi'\rangle$

a) $U_\varphi = |0\rangle\langle 0| + e^{i\varphi}|1\rangle\langle 1|$

$U_\varphi = \begin{bmatrix} 1 & 0 \\ 0 & e^{i\varphi} \end{bmatrix}$

$H|0\rangle = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}\begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}}\begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) = |+\rangle$

$U_\varphi|+\rangle = (|0\rangle\langle 0| + e^{i\varphi}|1\rangle\langle 1|)\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle)$

$= \frac{1}{\sqrt{2}}(|0\rangle\underbrace{\langle 0|0\rangle}_{1} + |0\rangle\underbrace{\langle 0|1\rangle}_{0} + e^{i\varphi}|1\rangle\underbrace{\langle 1|0\rangle}_{0} + e^{i\varphi}|1\rangle\underbrace{\langle 1|1\rangle}_{1})$

$= \frac{1}{\sqrt{2}}(|0\rangle + e^{i\varphi}|1\rangle)$

$H\left(\frac{1}{\sqrt{2}}|0\rangle + e^{i\varphi}|1\rangle\right) = \frac{1}{\sqrt{2}}(H|0\rangle + He^{i\varphi}|1\rangle) = \frac{1}{\sqrt{2}}(H|0\rangle + e^{i\varphi}H|1\rangle)$

$= \frac{1}{\sqrt{2}}(|+\rangle + e^{i\varphi}|-\rangle)$

$|\chi'\rangle = \frac{1}{\sqrt{2}}(|+\rangle + e^{i\varphi}|-\rangle)$

b) $|\chi'\rangle = \alpha|0\rangle + \beta|1\rangle$ $\quad \alpha^2 = p_0,\ \beta^2 = p_1$ and $\alpha^2 + \beta^2 = p_0 + p_1 = 1$

$|\chi'\rangle = \frac{1}{\sqrt{2}}\left(\frac{1}{\sqrt{2}}(|0\rangle + |1\rangle) + e^{i\varphi}\frac{1}{\sqrt{2}}(|0\rangle - |1\rangle)\right) = \frac{1}{2}((1 + e^{i\varphi})|0\rangle + (1 - e^{i\varphi}))$

$\alpha = \frac{1}{2}(1 + e^{i\varphi}) = \frac{1}{2}(1 + \cos\varphi + i\sin\varphi)$ $\quad \beta = \frac{1}{2}(1 - e^{i\varphi}) = \frac{1}{2}(1 - \cos\varphi - i\sin\varphi)$

$\alpha^2 = \frac{1}{4}(1 + \cos\varphi + i\sin\varphi + \cos\varphi + \cos^2\varphi + i\sin\varphi\cos\varphi - i\sin\varphi - i\sin\varphi\cos\varphi + \sin^2\varphi) = \frac{1}{4}(2 + 2\cos\varphi)$

$\beta^2 = \frac{1}{4}(1 - \cos\varphi + i\sin\varphi - \cos\varphi + \cos^2\varphi - i\sin\varphi\cos\varphi - i\sin\varphi + i\sin\varphi\cos\varphi + \sin^2\varphi) = \frac{1}{4}(2 - 2\cos\varphi)$

$p_0 = \frac{1 + \cos\varphi}{2}$ $\qquad p_1 = \frac{1 - \cos\varphi}{2}$

$p_0 - p_1 = \frac{1 + \cos\varphi - (1 - \cos\varphi)}{2}$ $\qquad p_0 - p_1 = \cos\varphi$

$$\boxed{\varphi = \cos^{-1}(p_0 - p_1)}$$

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2)

a) $R_y(\theta) = \cos\left(\frac{\theta}{2}\right) \mathbb{1} - i \sin\left(\frac{\theta}{2}\right) \sigma_y$

$$R_y(\theta) = \begin{bmatrix} \cos\theta/2 & 0 \\ 0 & \cos\theta/2 \end{bmatrix} - \begin{bmatrix} 0 & \sin\theta/2 \\ -\sin\theta/2 & 0 \end{bmatrix} = \begin{bmatrix} \cos\theta/2 & -\sin\theta/2 \\ \sin\theta/2 & \cos\theta/2 \end{bmatrix}$$

$R_z(\theta) = \cos\left(\frac{\theta}{2}\right) \mathbb{1} - i \sin\left(\frac{\theta}{2}\right) \sigma_z$

$$R_z(\theta) = \begin{bmatrix} \cos\frac{\theta}{2} & 0 \\ 0 & \cos\frac{\theta}{2} \end{bmatrix} - \begin{bmatrix} i\sin\theta/2 & 0 \\ 0 & -i\sin\theta/2 \end{bmatrix} = \begin{bmatrix} \cos\frac{\theta}{2} - i\sin\theta/2 & 0 \\ 0 & \cos\theta/2 + i\sin\theta/2 \end{bmatrix}$$

$$R_z(\theta) = \begin{bmatrix} e^{-i\theta/2} & 0 \\ 0 & e^{i\theta/2} \end{bmatrix}$$

$$|\chi'\rangle = R_z(\pi/2)\, R_y(\pi/2)\, |0\rangle$$

$$R_y(\pi/2) = \begin{bmatrix} \cos\pi/4 & -\sin\pi/4 \\ \sin\pi/4 & \cos\pi/4 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix}$$

$$R_z(\pi/2) = \begin{bmatrix} \cos\pi/4 - i\sin\pi/4 & 0 \\ 0 & \cos\pi/4 + i\sin\pi/4 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1-i & 0 \\ 0 & 1+i \end{bmatrix}$$

$$R_z(\pi/2)\, R_y(\pi/2) = \frac{1}{\sqrt{2}} \begin{bmatrix} 1-i & 0 \\ 0 & 1+i \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & -1 \\ 1 & 1 \end{bmatrix} = \frac{1}{2} \begin{bmatrix} 1-i & -1+i \\ 1+i & 1+i \end{bmatrix}$$

b) $|0\rangle \longrightarrow |+\rangle \longrightarrow |i\rangle$

$H|0\rangle = |+\rangle \qquad H = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix}$

$S|+\rangle = |i\rangle \qquad S = \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix}$

$|0\rangle$ — $\boxed{H}$ — $\boxed{S}$ — $|i\rangle$

$$\begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 1 \\ 1 & -1 \end{bmatrix} \begin{bmatrix} 1 \\ 0 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 & 0 \\ 0 & i \end{bmatrix} \begin{bmatrix} 1 \\ 1 \end{bmatrix} = \frac{1}{\sqrt{2}} \begin{bmatrix} 1 \\ i \end{bmatrix}$$

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3)

a) $R_x(\pi) = \begin{pmatrix} \cos\pi/2 & -i\sin\pi/2 \\ -i\sin\pi/2 & \cos\pi/2 \end{pmatrix} = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix}$

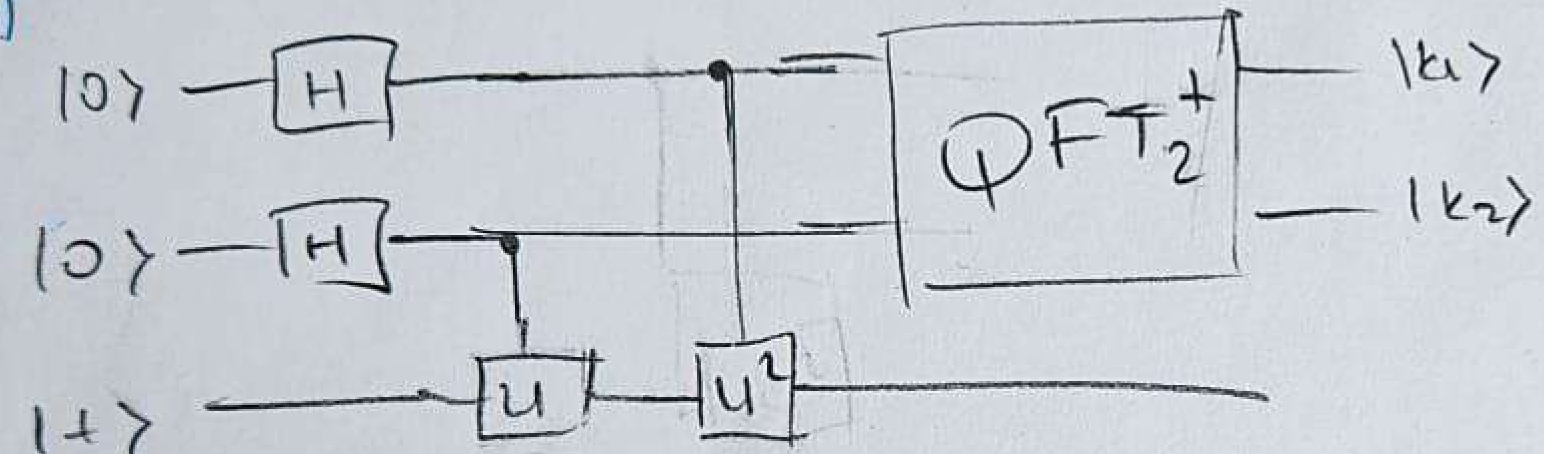
$U|+\rangle = \begin{pmatrix} 0 & -i \\ -i & 0 \end{pmatrix} \frac{1}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = \frac{1}{\sqrt{2}}\begin{pmatrix} -i \\ -i \end{pmatrix} = \frac{-i}{\sqrt{2}}\begin{pmatrix} 1 \\ 1 \end{pmatrix} = -i|+\rangle$

$U|+\rangle = \lambda|+\rangle \qquad \lambda = -i$

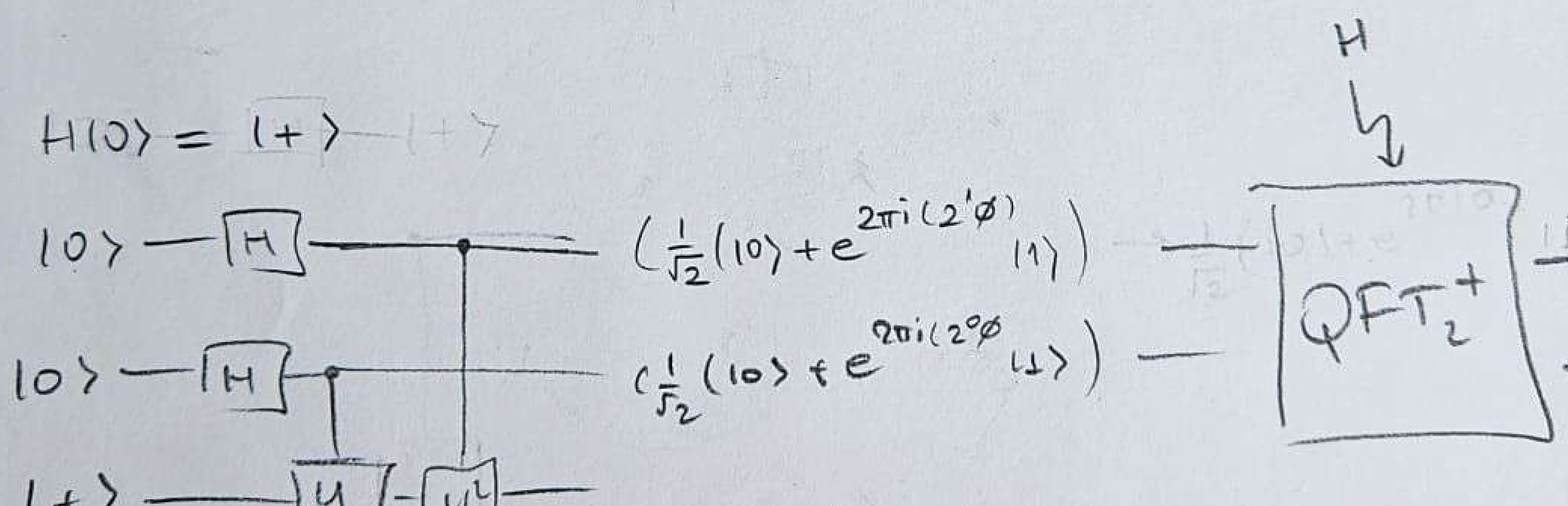
$\lambda = e^{2\pi i\phi} \qquad -i = e^{\frac{3\pi}{2}i} \qquad e^{\frac{3\pi}{2}i} = e^{2\pi i\phi}$

$\phi = \frac{3}{4}$ (decimal) $\qquad \phi = 0.11$ (binary) $\qquad k_1 = 1,\ k_2 = 1 \qquad |\phi\rangle = |k_1 k_2\rangle = |11\rangle$

b)



$H|0\rangle = |+\rangle$



$\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i(2.(0.11))}|1\rangle) = \frac{1}{\sqrt{2}}(|0\rangle + \cancel{e^{2\pi i}}^{\,1} . e^{2\pi i\, 0.1}|1\rangle)$

$e^{2\pi i \cdot \frac{1}{2}} = e^{\pi i} = -1$

- $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i.\,0.1}|1\rangle) \xrightarrow{H\otimes 1} \frac{1}{\sqrt{2}}(|+\rangle + e^{2\pi i.\,0.1}|-\rangle) = \frac{1}{2}(|0\rangle + |1\rangle + \underbrace{e^{2\pi i\,0.1}}_{-1}(|0\rangle - |1\rangle)$

$= \frac{1}{2}(|0\rangle + |1\rangle + (-1)|0\rangle - |1\rangle) = \frac{1}{2}(2|1\rangle) = |1\rangle$

$e^{2\pi i\cdot\frac{3}{4}} = e^{\frac{3\pi}{2}i} = (-i)$

- $\frac{1}{\sqrt{2}}(|0\rangle + e^{2\pi i\,0.11}|1\rangle) \xrightarrow{H\otimes 1} \frac{1}{\sqrt{2}}(|+\rangle + e^{2\pi i.\,0.11}|-\rangle) = \frac{1}{2}(|0\rangle + |1\rangle + e^{2\pi i.\frac{0.11}{4}}(|0\rangle - |1\rangle)$

$= \frac{1}{2}(|0\rangle + |1\rangle + (-i)\ |0\rangle - |1\rangle) = \frac{1}{2}\left(\begin{bmatrix} 1 \\ 1 \end{bmatrix} + \begin{bmatrix} -i \\ +i \end{bmatrix}\right) = \frac{1}{2}\begin{pmatrix} 1-i \\ 1+i \end{pmatrix}$

$\underline{\underline{|k_1\rangle = |1\rangle}} \qquad \underline{\underline{|k_2\rangle = \frac{1}{2}\begin{bmatrix} 1-i \\ 1+i \end{bmatrix}}}$